Harisankar S

DEEP LEARNING RESEARCHER • VISION SYSTEMS

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// EDUCATION

TKM College of Engineering (TKMCE) 2023 – 2027 (Expected)
B.Tech in Computer Science & Engineering Kollam, Kerala

Maintaining a cumulative CGPA of 8.90 / 10. Core coursework: Data Structures, Operating Systems, Computer Networks, Database Management Systems, and Theory of Computation.

Christ Nagar Central School 2023
Higher Secondary (Class XII) - Science Thiruvananthapuram, Kerala

Graduated with 94% in Science stream with Mathematics and Computer Science.

St. Thomas Residential School 2021
Secondary School (Class X) Thiruvananthapuram, Kerala

Graduated with 96% academic distinction.

// RESEARCH & INTERNSHIP EXPERIENCE

Deep Learning Research Intern May 2026 – Jul 2026
National Institute of Technology Calicut (NIT Calicut) — Calicut, Kerala

Conducted medical vision research in the Dept. of CSE. Developed the Wavelet Dual-Stream Hybrid Restormer on NVIDIA H200 GPUs, achieving 37.39 dB PSNR on ultrasound datasets.

Computer Vision / Deep Learning Intern Jun 2025 – Jul 2025
Kerala State Remote Sensing and Environment Centre (KSREC) — Thiruvananthapuram, Kerala

Built Siamese Swin Transformer architectures for pixel-level land-use change detection on the LEVIR-CD benchmark with 96.9% accuracy to automate state GIS environmental monitoring.

// LEADERSHIP & STUDENT COMMUNITY

Chairperson, ACM Student Chapter Jun 2026 – Present
TKM College of Engineering — Kollam, Kerala

Elected to lead the executive committee, directing the vision, technical hackathons, research reading groups, and developer workshops for 500+ undergraduate student members.

// SELECTED PROJECTS & CLOUD ARCHITECTURE

Wavelet Dual-Stream Hybrid Restormer May 2026 – Jul 2026

Medical Deep Learning & Image Denoising • Tech: PyTorch, Vision Transformers, Restormer, EdgeCNN, Wavelet Transform, NVIDIA H200, Albumentations

Engineered a 26M-parameter Vision Transformer combined with a custom EdgeCNN to eliminate musculoskeletal ultrasound speckle noise while preserving diagnostic tissue boundaries. Trained with a progressive patch-size curriculum (128px → 384px) on NVIDIA H200 GPUs.

Result: **37.39 dB** (Fidelity PSNR)

Siamese Swin Transformer for Satellite Change Detection Jun 2025 – Jul 2025

Geospatial Computer Vision & Remote Sensing • Tech: Python, PyTorch, Swin Transformer, OpenCV, LEVIR-CD, QGIS, NumPy

Built a Siamese Swin Transformer (SwinV2-Tiny backbone) with differential feature fusion and a convolutional decoder to detect pixel-level urban expansion and deforestation across Kerala from bi-temporal satellite imagery.

Result: **96.9%** (Pixel Accuracy)

Hop On — Concurrent Cab Booking System 2024

Full-Stack Software Architecture • Tech: Java, JavaFX, OOP Design Patterns, Concurrency, State Machines

Architected a multi-module desktop cab booking platform in Java with separate concurrent workflows for riders and drivers. Designed thread-safe dispatch algorithms, user authentication, and a responsive JavaFX graphical interface.

Result: **Decoupled MVC** (Architecture)

// TECHNICAL TOOLKIT

CLOUD & INFRASTRUCTURE (AWS):

AWS Amplify Hosting, Amazon CloudFront, Amazon S3, AWS Lambda (Serverless), Route 53 & ACM

DEEP LEARNING & COMPUTER VISION:

PyTorch, Vision Transformers (Restormer, Swin), Wavelet Transforms, OpenCV & Albumentations, Convolutional Neural Networks (CNN)

LANGUAGES & COMPUTER SCIENCE CORE:

Python, TypeScript / JavaScript, Java, C, Data Structures & Algorithms

// FOUNDATIONAL HONORS

National 2nd Runner-Up (Mental Arithmetic) — National Abacus Championship (National Level)
Secured National 2nd Runner-Up and Kerala State 1st Runner-Up in high-speed mental arithmetic, establishing deep foundational numerical and spatial visualization discipline.